The content-(1^n) census of the Warnaar determinant is a simple random walk, and the t-degree of $c_{(1^n)}$ grows quadratically at every k

Clio

2026-10-14

PROTOCOL §2.3. **Author:** Clio (prove session, 2026-10-14 c2). **Recipient:** Robin Langer; cc Rick. **Title:** as above. **Source commit:** `a83a008` in `clio-vega/proofs`. **Questions:** Q458 (Theorem 3.5), Q457 (Theorem 4.4). **Code:** `proofs/code-1014-q458/`.

Abstract

Let G_t be Warnaar's determinant (`Eq_generalise`) of size k with parameters $M = k+\ell+1$, $N = 2M$, and let $c_\alpha(t)$ be the coefficient of x^α in $G_t \cdot (x_1 \cdots x_n)^k$. We compute the fibre $\alpha = (1^n)$ exactly.

The computation rests on one observation, which we call the *census*: for the constant target exponent that $\alpha = (1^n)$ forces, the n variables decouple, and the resulting local generating function is $\big(\prod_i z_i\big) \sum_{i=1}^{k} (z_i + z_i^{-1})$. Hence the x-side of the Warnaar determinant at content (1^n) **is the simple random walk on** $\mathbb{Z}^k$ (Theorem 2.2). Two consequences follow.

First (Theorem 3.5, **Q458**), at $k = 1$ the whole fibre collapses to a ballot-number sum and $\deg_t c_{(}1^n)(t) = \max\{My^2 - y : y \in \mathbb{Z},\ -n/N \le y \le (n+2)/N\}$ for every even n, with $c_{(}1^n) \equiv 0$ for odd n. This closes the single non-vacuity gap left open by `2026-10-14-Q452-transfer-matrix-capacity.tex`: in fact *every* admissible winding sector contributes a nonzero coefficient, not merely the extremal one.

Second (Theorem 4.4, **Q457**), at *every* $k \ge 1$ the permutation index σ cancels out of the t-exponent after completing the square, and the maximisation becomes a convex maximum over an ℓ^1 ball with a unique vertex. Consequently, whenever $N \mid n$ or $N \mid n + 2k$,

$$\deg_t c_{(}1^n)(t) = \frac{n(n+2k)}{4M}, \qquad \text{leading coefficient} \quad +1 \text{ resp. } -1.$$

Since each residue class is infinite, $\deg_t c_{(}1^n)$ is $\Theta(n^2)$ for every (k, ℓ). No weight whose value on a content-(1^n) cylindric tableau is a product of n polynomials of degree $\le d$ can satisfy Warnaar's identity for all n (Corollary 4.6) — **including time-dependent weights**, which is the repair that Specification 6 of the Q452 paper proposed and which is therefore dead before it was attempted.

All statements are verified against two engines sharing no code path: the sympy symmetric-function determinant engine M1 (`proofs/code-1010-q410`) and a determinant-free census engine written for this paper. **Scope is stated in §5 and is not what one would guess.**

1 Setting and conventions

Fix $n \ge 1$, $k \ge 1$, $\ell \ge 0$, and put

$$M := k + \ell + 1, \qquad N := 2M, \qquad h := 2k, \qquad w := 2\ell + 2,$$

so that $N = 2k + 2\ell + 2 = h + w$. Let $x = (x_1, \dots, x_n)$ and

$$\dot{e}_r(x) := e_r\big(x_1, x_1^{-1}, \dots, x_n, x_n^{-1}\big), \qquad 0 \le r \le 2n,$$

with $\dot{e}_r := 0$ outside that range. Warnaar's determinant (`Eq_generalise`, transcribed verbatim in `code-1010-q410/core.py`) is

$$G_t \;=\; \sum_{y \in \mathbb{Z}^k} \det_{1 \le i,j \le k} \Big(t^{My_i^2 - j\,y_i} \big(\dot{e}_{\,n-i+j-Ny_i}(x) - \dot{e}_{\,n+i+j-Ny_i}(x) \big) \Big), \tag{1}$$

and for a composition α of n,

$$c_\alpha(t) \;:=\; [x^\alpha]\, \big(G_t \cdot (x_1 \cdots x_n)^k \big). \tag{2}$$

Throughout we study the fibre $\alpha = (1^n)$.

Remark 1.1 (two different h's). The determinant size in (1) is k; the cylindric parameter in Korff's and in HKKO's conventions is $h = 2k$. We never write h for the determinant size.

2 The census: the x-side is a simple random walk

Lemma 2.1 (reduction to a constant target exponent)**.** *$c_{(1^n)}(t) = [x^{(1-k)\cdot(1,\dots,1)}]\, G_t$, the coefficient of the Laurent monomial $(x_1 \cdots x_n)^{1-k}$ in G_t. In particular at $k = 1$ it is the constant term of G_t.*

Proof. Immediate from (2): $[x^{(1^n)}](G_t \cdot (x_1 \cdots x_n)^k) = [x^{(1^n) - k(1^n)}] G_t$. □

The point of Lemma 2.1 is that the target exponent vector $b \cdot (1, \dots, 1)$ with $b := 1 - k$ is *constant in the variable index*. That is what makes the next lemma a product formula.

Theorem 2.2 (census)**.** *Let $b = 1 - k$ and let $r_1, \dots, r_k \in \mathbb{Z}$. Write*

$$W_n(p_1, \dots, p_k) \;:=\; \#\Big\{ \text{n-step walks on } \mathbb{Z}^k \text{ with steps } \pm e_i \text{ from } 0 \text{ to } (p_1, \dots, p_k) \Big\}.$$

Then

$$[x^{\,b\cdot(1,\dots,1)}] \prod_{i=1}^{k} \dot{e}_{r_i}(x) \;=\; W_n\big(r_1 - n,\; \dots,\; r_k - n\big). \tag{3}$$

Equivalently, $\sum_r \big([x^{b(1,\dots,1)}] \prod_i \dot{e}_{r_i} \big) z^r = \big(\prod_i z_i \big)^n \big(\sum_{i=1}^k (z_i + z_i^{-1}) \big)^n$.

Proof. Expand each $\dot{e}_{r_i} = e_{r_i}$ over subsets of the $2n$ distinct formal variables $\{x_j^{\pm 1}\}$. A term of $\prod_i \dot{e}_{r_i}$ is a choice, for each factor i and each variable index j, of a subset $S_{i,j} \subseteq \{x_j, x_j^{-1}\}$; it contributes $|S_{i,j}|$ to r_i and $\varepsilon(S_{i,j}) := [x_j \in S_{i,j}] - [x_j^{-1} \in S_{i,j}] \in \{-1, 0, 1\}$ to the exponent of x_j. The four local states are

$$\emptyset : (\varepsilon, \text{size}) = (0, 0), \quad \{x_j\} : (1, 1), \quad \{x_j^{-1}\} : (-1, 1), \quad \{x_j, x_j^{-1}\} : (0, 2).$$

The constraint is $\sum_i \varepsilon(S_{i,j}) = b$ for each j separately, so the n variables decouple and the full generating function (tracking r_i by z_i) is $L_b(z)^n$ with

$$L_b(z) \;=\; \sum_{\substack{\varepsilon \in \{-1,0,1\}^k \\ \sum_i \varepsilon_i = b}} \prod_i \big(z_i \text{ if } \varepsilon_i \neq 0, \;\; 1 + z_i^2 \text{ if } \varepsilon_i = 0 \big).$$

Now count: writing $u = \#\{\varepsilon_i = 1\}$, $v = \#\{\varepsilon_i = -1\}$, $z = \#\{\varepsilon_i = 0\}$, the two conditions $u - v = b = 1 - k$ and $u + v + z = k$ give $z = 1 - 2u$, hence $u = 0$, $z = 1$, $v = k - 1$. *Exactly one coordinate carries $\varepsilon_i = 0$ and all the others carry $\varepsilon_i = -1$.* Therefore

$$L_{1-k}(z) \;=\; \sum_{i=1}^{k}(1+z_i^2)\prod_{j\neq i} z_j \;=\; \Big(\prod_j z_j\Big)\sum_{i=1}^{k}\big(z_i + z_i^{-1}\big). \tag{4}$$

Raising (4) to the n-th power and substituting $p_i = r_i - n$ gives (3), because $\big(\sum_i (z_i + z_i^{-1})\big)^n$ is by definition the generating function of n-step $\pm e_i$ walks on $\mathbb{Z}^k$. □

Remark 2.3 (reading of Theorem 2.2). Combinatorially the proof says: at content (1^n), each of the n variables must, across the k determinant factors, be "half-used" in exactly $k-1$ of them (state $\{x_j^{-1}\}$) and "either unused or fully used" in the remaining one. The choice of which factor is the exceptional one, together with the unused/fully-used bit, is precisely a choice of a coordinate direction and a sign — one step of a walk on $\mathbb{Z}^k$. The Warnaar determinant does not see the partition structure of the fibre at all; it sees n steps of a walk.

Corollary 2.4 (support). *$W_n(p) \neq 0$ if and only if $\|p\|_1 \leq n$ and $\|p\|_1 \equiv n \pmod 2$. At $k = 1$, $W_n(p) = \binom{n}{(n+p)/2}$, so*

$$[x^0]\,\dot{e}_r \;=\; \binom{n}{r/2} \quad (r \text{ even}), \qquad 0 \quad (r \text{ odd}). \tag{5}$$

3 Theorem A: the exact degree law at $k = 1$ (Q458)

At $k = 1$ the determinant (1) is 1×1 and

$$G_t = \sum_{y\in\mathbb{Z}} t^{My^2 - y}\big(\dot{e}_{\,n-Ny} - \dot{e}_{\,n+2-Ny}\big), \qquad M = \ell + 2. \tag{6}$$

Lemma 3.1 (closed form). *If n is odd then $c_(1^n)(t) \equiv 0$. If $n = 2m$ is even then*

$$c_(1^n)(t) \;=\; \sum_{y\in\mathbb{Z}} t^{\,My^2 - y}\, D(m - My), \qquad D(j) := \binom{2m}{j} - \binom{2m}{j+1}. \tag{7}$$

Proof. By Lemma 2.1 and (6), $c_(1^n) = [x^0]G_t = \sum_y t^{My^2-y}\big([x^0]\dot{e}_{n-Ny} - [x^0]\dot{e}_{n+2-Ny}\big)$. Since $N = 2M$ is even, both $n - Ny$ and $n + 2 - Ny$ have the parity of n; by (5) both constant terms vanish when n is odd. For $n = 2m$, (5) gives $\binom{n}{m-My}$ and $\binom{n}{m+1-My}$. □

Lemma 3.2 (sign and support of the ballot difference). *For $m \geq 0$ and $j \in \mathbb{Z}$:*

$$D(j) > 0 \iff m \leq j \leq 2m, \qquad D(j) < 0 \iff -1 \leq j \leq m-1, \qquad D(j) = 0 \iff j \leq -2 \text{ or } j \geq 2m+1.$$

In particular $D(j) \neq 0$ if and only if $-1 \leq j \leq 2m$, and $D(m) = \mathrm{Cat}_m > 0$.

Proof. $\binom{2m}{\cdot}$ vanishes outside $[0, 2m]$, is strictly increasing on $[0, m]$ and strictly decreasing on $[m, 2m]$. Hence: $j \geq 2m+1$ gives $0 - 0$; $j = 2m$ gives $1 - 0 > 0$; $m \leq j \leq 2m-1$ gives $\binom{2m}{j} > \binom{2m}{j+1}$; $0 \leq j \leq m-1$ gives $\binom{2m}{j} < \binom{2m}{j+1}$; $j = -1$ gives $0 - \binom{2m}{0} = -1 < 0$; $j \leq -2$ gives $0 - 0$. ($m = 0$: the middle range $[0, m-1]$ is empty and $j = 0$ falls in $[m, 2m] = \{0\}$ with $D(0) = 1$.) The value $D(m) = \binom{2m}{m} - \binom{2m}{m+1}$ is the m-th Catalan number. □

Lemma 3.3 (the support is exactly the admissible range)**.** *Let $n = 2m$. Then $D(m - My) \neq 0$ if and only if*

$$-\frac{n}{N} \;\le\; y \;\le\; \frac{n+2}{N}.$$

Proof. By Lemma 3.2, $D(m - My) \neq 0 \iff -1 \le m - My \le 2m \iff -m \le My \le m + 1$. Doubling and using $N = 2M$, $n = 2m$: $-n \le Ny \le n + 2$. □

Lemma 3.4 (no cross-sector cancellation)**.** *For $M \ge 2$ the map $\varphi(y) = My^2 - y$ is injective on $\mathbb{Z}$.*

Proof. $\varphi(y_1) = \varphi(y_2)$ gives $(y_1 - y_2)\big(M(y_1 + y_2) - 1\big) = 0$. If $y_1 \neq y_2$ then $M(y_1 + y_2) = 1$ with $M \ge 2$ and $y_1 + y_2 \in \mathbb{Z}$, impossible. □

Theorem 3.5 (Q458; the $k = 1$ degree law)**.** *Let $k = 1$, $\ell \ge 0$, $M = \ell + 2$, $N = 2M$. Then $c_{(}1^n)(t) \equiv 0$ for odd n, and for even n,*

$$\boxed{\deg_t c_{(}1^n)(t) \;=\; \max\Big\{\, My^2 - y \;:\; y \in \mathbb{Z},\; -\tfrac{n}{N} \le y \le \tfrac{n+2}{N} \,\Big\}.}$$

Moreover $c_{(}1^n)$ has exactly $\#\{y \in \mathbb{Z} : -n/N \le y \le (n+2)/N\}$ nonzero terms, its constant term is $\mathrm{Cat}_{n/2} > 0$, and the coefficient of t^{My^2-y} is positive for $y \le 0$ and negative for $y \ge 1$.

Proof. Odd n is Lemma 3.1. Let $n = 2m$. By Lemma 3.1, $c_{(}1^n) = \sum_{y\in\mathbb{Z}} D(m - My)\, t^{\varphi(y)}$. By Lemma 3.3 the set of y with $D(m - My) \neq 0$ is exactly $Y(n) := \{y \in \mathbb{Z} : -n/N \le y \le (n+2)/N\}$. By Lemma 3.4 the exponents $\varphi(y)$, $y \in Y(n)$, are pairwise distinct, so no two surviving terms can combine: the polynomial has exactly $|Y(n)|$ nonzero terms and its degree is $\max_{y\in Y(n)} \varphi(y)$. The sign statement is Lemma 3.2 with $j = m - My$: $y \le 0 \Rightarrow j \ge m$ and $y \ge 1 \Rightarrow j \le m - M \le m - 2 < m$ while $j \ge -1$ by admissibility. Finally $0 \in Y(n)$ always, contributing $D(m) = \mathrm{Cat}_m$ at exponent 0. □

Remark 3.6 (what exactly was the gap, and what closed it). The gap recorded in `2026-10-14-Q452-transfer-matr` was *(G): the extremal admissible y contributes a nonzero coefficient.* What Lemma 3.3 proves is strictly stronger and strictly easier: *every* admissible y does. The reason the stronger statement is the easier one is that it is a statement about a *set*, decided by a single monotonicity of binomial coefficients, whereas (G) as posed is a statement about an *extremum*, which invites a search for the extremum first. The instrument was the census, run before any mechanism: once $[x^0]\dot{e}_r$ is named as $\binom{n}{r/2}$, there is nothing left to do.

Proposition 3.7 (explicit evaluation of the maximum)**.** *Let n be even, $A := \lfloor n/N \rfloor$ and $B := \lfloor (n+2)/N \rfloor \in \{A, A+1\}$. Then*

$$\deg_t c_{(}1^n) = \begin{cases} MB^2 - B, & B = A + 1, \\ MA^2 + A, & B = A. \end{cases}$$

Consequently $\deg_t c_{(}1^n) \;\ge\; MA^2 - A \;\ge\; \dfrac{n^2}{4M} - \dfrac{3}{2}n$ *for all even $n \ge N + 2$, and* $\deg_t c_{(}1^n) = \dfrac{n^2}{4M} + O(n)$.

Proof. $\varphi(y) = My^2 - y$ is convex with vertex $1/(2M) \in (0, \frac{1}{2})$, so on the integer interval $[-A, B]$ its maximum is at an endpoint: $\max(\varphi(-A), \varphi(B)) = \max(MA^2 + A,\ MB^2 - B)$. If $B = A$ then $MA^2 + A \geq MA^2 - A$. If $B = A+1$ then $\varphi(A+1) - \varphi(-A) = M(2A+1) - (A+1) - A = (M-1)(2A+1) > 0$ since $M \geq 2$. For the bound: $\varphi(A+1) \geq \varphi(A) = MA^2 - A$ and $\varphi(-A) \geq MA^2 - A$, so in both cases $\deg \geq MA^2 - A = A(MA-1)$. With $A \geq n/N - 1 = n/(2M) - 1$ and $n \geq N + 2$ both factors are positive, and $A(MA-1) \geq \left(\frac{n}{2M} - 1\right)\left(\frac{n}{2} - M - 1\right) = \frac{n^2}{4M} - \frac{2M+1}{2M} n + M + 1 \geq \frac{n^2}{4M} - \frac{3}{2} n$, using $\frac{2M+1}{2M} \leq \frac{5}{4} < \frac{3}{2}$ for $M \geq 2$. The upper bound $\deg \leq M \max(A, B)^2 + \max(A, B) \leq n^2/(4M) + O(n)$ is immediate from $A, B \leq (n+2)/N$. □

4 Theorem B: quadratic degree at every k (Q457)

Expanding the determinant (1) and the k binomial differences, and applying Theorem 2.2, gives a finite sum with no symmetric functions left in it. Fix the data

$$y \in \mathbb{Z}^k, \qquad \sigma \in S_k, \qquad \varepsilon \in \{\pm 1\}^k,$$

where $\varepsilon_i = +1$ selects the term $-\dot{e}_{n+i+\sigma(i)-Ny_i}$ and $\varepsilon_i = -1$ selects $+\dot{e}_{n-i+\sigma(i)-Ny_i}$. Put

$$p_i \ := \ \sigma(i) + \varepsilon_i\, i - N y_i \qquad (1 \leq i \leq k), \tag{8}$$

so that $r_i = n + p_i$ is the index of the $\dot{e}$ chosen in row i. Then

$$c_{(1^n)}(t) \ = \ \sum_{y \in \mathbb{Z}^k} \ \sum_{\sigma \in S_k} \ \sum_{\varepsilon \in \{\pm 1\}^k} \operatorname{sgn}(\sigma) \left(\prod\nolimits_i (-\varepsilon_i) \right) t^{\,E(y,\sigma)}\ W_n(p), \qquad E(y,\sigma) := \textstyle\sum_i \big(M y_i^2 - \sigma(i) y_i \big), \tag{9}$$

where $\prod_i (-\varepsilon_i) = (-1)^{\#\{i : \varepsilon_i = +1\}}$.

Lemma 4.1 (completing the square: σ cancels)**.** *For every (y, σ, ε) with p as in (8),*

$$E(y,\sigma) \ = \ \frac{1}{4M} \Big[\ \sum_{i=1}^{k} \big(p_i - \varepsilon_i\, i \big)^2 \ - \ \sum_{s=1}^{k} s^2 \ \Big]. \tag{10}$$

In particular E depends on (y, σ, ε) only through the vector $\big(p_i - \varepsilon_i i\big)_i = \big(-N y_i + \sigma(i)\big)_i$, and the permutation enters only through the constant $\sum_s s^2$.

Proof. Write $q_i := N y_i = 2M y_i$. Then $E = M \sum q_i^2/(4M^2) - \sum \sigma(i) q_i/(2M) = \frac{1}{4M}\big[\sum q_i^2 - 2 \sum \sigma(i) q_i\big] = \frac{1}{4M}\big[\sum (q_i - \sigma(i))^2 - \sum \sigma(i)^2\big]$. Since σ is a permutation of $\{1, \dots, k\}$, $\sum_i \sigma(i)^2 = \sum_{s=1}^{k} s^2$. Finally (8) gives $q_i - \sigma(i) = \varepsilon_i i - p_i$. □

Lemma 4.2 (the convex maximum and its unique vertex)**.** *Let $k, n \geq 1$. Over all $p \in \mathbb{Z}^k$ with $\|p\|_1 \leq n$ and all $\varepsilon \in \{\pm 1\}^k$,*

$$\sum_{i=1}^{k} \big(p_i - \varepsilon_i i \big)^2 \ \leq \ (n+k)^2 + \sum_{s=1}^{k-1} s^2,$$

with equality if and only if $|p_k| = n$, $p_i = 0$ for $i < k$, and $\varepsilon_k = -\operatorname{sgn}(p_k)$ (the values of ε_i for $i < k$ being immaterial to the left-hand side).

Proof. Put $b_i := |p_i| \geq 0$, so $\sum b_i \leq n$. For each i, $(p_i - \varepsilon_i i)^2 \leq (b_i + i)^2$, with equality iff $\varepsilon_i = -\operatorname{sgn}(p_i)$ (and automatically when $b_i = 0$). So it suffices to maximise $\sum_i (b_i + i)^2 = \sum_i b_i^2 + 2\sum_i i\, b_i + \sum_s s^2$ over $b \in \mathbb{Z}^k_{\geq 0}$, $\sum b_i \leq n$.

Let $B := \sum b_i \leq n$. Then $\sum b_i^2 \leq B^2$, with equality iff b is supported on a single coordinate; and $2\sum i\, b_i \leq 2kB$, with equality iff b is supported on $\{k\}$. Hence $\sum b_i^2 + 2\sum i b_i \leq B^2 + 2kB \leq n^2 + 2kn$, the last step because $B \mapsto B^2 + 2kB$ is increasing on $B \geq 0$. Equality throughout forces $B = n$ and $b = n\, e_k$, i.e. $|p_k| = n$ and $p_i = 0$ for $i < k$; and then $\sum_i (p_i - \varepsilon_i i)^2 = (n+k)^2 + \sum_{s<k} s^2$ exactly when $\varepsilon_k = -\operatorname{sgn}(p_k)$ (the other sign gives $(n-k)^2 < (n+k)^2$). □

Lemma 4.3 (which vertices are realisable)**.** *Let $n \geq 1$ and suppose (y, σ, ε) realises the equality case of Lemma 4.2, i.e. $p = \pm n\, e_k$ with $\varepsilon_k = -\operatorname{sgn}(p_k)$ and $p_i = 0$ for $i < k$. Then necessarily $\sigma = \mathrm{id}$ and $\varepsilon_i = -1$ for all $i < k$, and exactly one of the following holds:*

(B) *$p = +n\, e_k$, $\varepsilon = (-1, \ldots, -1)$, $y = -\frac{n}{N} e_k$ — possible if and only if $N \mid n$; the signed contribution of this term to (9) is $+1$;*

(A) *$p = -n\, e_k$, $\varepsilon = (-1, \ldots, -1, +1)$, $y = \frac{n+2k}{N} e_k$ — possible if and only if $N \mid n + 2k$; the signed contribution is -1.*

Cases (A) and (B) are mutually exclusive.

Proof. Realisability means $y_i = \big(\sigma(i) + \varepsilon_i i - p_i\big)/N \in \mathbb{Z}$ for every i.

Coordinates $i < k$. Here $p_i = 0$, so we need $N \mid \sigma(i) + \varepsilon_i i$. Since $1 \leq \sigma(i) \leq k$ and $1 \leq i \leq k-1$ we have $0 < \sigma(i) + i \leq 2k - 1 < 2k + 2 \leq N$, so the choice $\varepsilon_i = +1$ is impossible; and $|\sigma(i) - i| \leq k - 1 < N$, so $\varepsilon_i = -1$ forces $\sigma(i) = i$ and $y_i = 0$. Thus σ fixes $1, \ldots, k-1$, hence $\sigma = \mathrm{id}$ and $\sigma(k) = k$.

Coordinate k. We need $N \mid k + \varepsilon_k k - p_k$. If $p_k = +n$ then $\varepsilon_k = -1$ and the condition is $N \mid -n$, i.e. $N \mid n$, giving $y_k = -n/N$; the sign in (9) is $\operatorname{sgn}(\mathrm{id}) \cdot (-1)^0 = +1$ and $W_n(n e_k) = 1$ (the walk must take all n steps in the direction $+e_k$), so the contribution is $+1$. If $p_k = -n$ then $\varepsilon_k = +1$ and the condition is $N \mid 2k + n$, giving $y_k = (n+2k)/N$; the sign is $(+1) \cdot (-1)^1 = -1$ and $W_n(-n e_k) = 1$, so the contribution is -1. If both held then $N \mid 2k$, impossible since $0 < 2k < 2k + 2 \leq N$. □

Theorem 4.4 (Q457; quadratic degree at every k)**.** *Let $k \geq 1$, $\ell \geq 0$, $M = k + \ell + 1$, $N = 2M$, and let $n \geq 1$ satisfy*

$$N \mid n \qquad \text{or} \qquad N \mid n + 2k.$$

Then

$$\boxed{\deg_t c_(1^n)(t) \;=\; \frac{n(n+2k)}{4M}},$$

and the leading coefficient is $+1$ in the first case and -1 in the second. In particular, for every fixed (k, ℓ) the degree is $\Theta(n^2)$ along each of these two infinite arithmetic progressions.

Proof. By (9) every contributing triple (y, σ, ε) has $W_n(p) \neq 0$, hence $\|p\|_1 \leq n$ by Corollary 2.4. By Lemma 4.1 its exponent is $E = \frac{1}{4M}\big[\sum_i (p_i - \varepsilon_i i)^2 - \sum_{s \leq k} s^2\big]$, and by Lemma 4.2 this is at most $\frac{1}{4M}\big[(n+k)^2 + \sum_{s<k} s^2 - \sum_{s \leq k} s^2\big] = \frac{1}{4M}\big[(n+k)^2 - k^2\big] = \frac{n(n+2k)}{4M}$, with equality only at the vertices described there.

By Lemma 4.3, under the hypothesis $N \mid n$ (resp. $N \mid n + 2k$) exactly one triple (y, σ, ε) realises a maximising vertex, and its signed contribution to (9) is $+1$ (resp. -1). Every other contributing triple has strictly smaller E. Hence the coefficient of $t^{n(n+2k)/(4M)}$ in $c_(1^n)$ is $\pm 1 \neq 0$ and no term of higher degree occurs. □

Remark 4.5 (why σ had to cancel). Lemma 4.1 is the whole content of Theorem 4.4. Before it, the maximisation is over $\mathbb{Z}^k \times S_k \times \{\pm 1\}^k$ with a constraint coupling all three; after it, the objective is a convex function of p alone on an ℓ^1 ball and the permutation has been absorbed into an additive constant. The ℓ^1 ball is the reason the answer is $n^2/(4M)$ for *every* k rather than $k \cdot n^2/(4M)$: a convex maximum over ℓ^1 concentrates on one coordinate, so raising k buys no extra winding. That the walk constraint is ℓ^1 and not a box is Theorem 2.2.

Corollary 4.6 (capacity: no bounded-degree multiplicative weight)**.** *Fix $k \geq 1, \ell \geq 0$ and $d \geq 0$. Suppose a weight assigns to each cylindric tableau T of content (1^n) a polynomial $W_T(t) = \prod_{a=1}^{n} \omega_a(T)(t)$ with $\deg_t \omega_a(T) \leq d$ for every a and T. Then*

$$\sum_{T \in \mathcal{T}(k,\ell,(1^n))} c^-(T)\, W_T(t) \;=\; c_{(}1^n)(t)$$

fails for all sufficiently large n in either progression of Theorem 4.4: explicitly, for all such $n > 4Md + 2k$.

Proof. The left side has degree at most $\max_T \deg_t W_T \leq nd$. By Theorem 4.4 the right side has degree $n(n+2k)/(4M) > nd$ as soon as $n + 2k > 4Md$. □

Remark 4.7 (the class this kills). The only hypothesis in Corollary 4.6 is that each of the n local factors has degree $\leq d$. The factor $\omega_a(T)$ may depend on the step index a, on the entire history $x^{(0)}, \ldots, x^{(a-1)}$, on the full multiplicity vector, on anything at all. So the corollary kills, simultaneously:

- multiplicity-determined weights (M), index-aware weights (I) and configuration-determined weights (C) of Definition 5 of the Q452 paper, drawn from a finite set of polynomials;
- **time-dependent** weights — i.e. exactly the repair proposed by Specification 6 of the Q452 paper, *before that repair was attempted*;
- at **every** (k, ℓ), not only at $(1, 0)$ and $(1, 1)$.

What it does **not** kill is a weight whose local factors grow in degree with n; a family ω depending on n with $\deg \omega_a = \Theta(n)$ is untouched, and indeed Warnaar's own identity is such an object.

5 Scope — stated before it can be overstated

1. **Theorem 3.5 is a $k = 1$ statement, hence $h = 2k = 2$, hence *outside* Korff's 2011 domain $h > 2$.** It is a statement about Warnaar's identity, which holds for all $k \geq 1$; it is not a statement about Korff's construction.

2. **Theorem 4.4 holds at every k, hence $h = 2k$ can exceed 2 — but it needs $n \to \infty$, and Korff's domain also requires $n \leq w = 2\ell + 2$.** Any asymptotic refutation necessarily leaves that finite window. So Corollary 4.6 refutes the weight class *as a candidate for Warnaar's identity at all n*; it does not refute anything inside Korff's finite window, where there is no asymptotics to be had. This is the complement of the Q452 result, which was finite-n but only at $(1, 0)$ and $(1, 1)$. The two together leave open exactly one region: $h > 2$ *and* $n \leq w$.

3. Theorem 4.4 covers only n in the two residue classes 0 and $-2k$ modulo N. For other n the maximising vertex of Lemma 4.2 is not realisable and the degree is strictly smaller; the measured values there (see §6) show several configurations tying at the top with a larger leading coefficient. The exact law for those n is **not** proved here. At $k = 1$ it is — Theorem 3.5 covers all even n — and the two theorems agree where they overlap.

4. Nothing here concerns fibres $\alpha \neq (1^n)$. Theorem 2.2 is specific to (1^n): it is exactly the constancy of the target exponent vector that makes the n variables decouple.

6 Computational verification

Two engines sharing no code path.

M1 `proofs/code-1010-q410/{core,warn}.py`: builds e_r of $2n$ sympy symbols by the elementary-symmetric recursion, forms the $k \times k$ determinant (1) symbolically, multiplies by $(x_1 \cdots x_n)^k$ and extracts $[x^{(1^n)}]$ variable by variable. Written 2026-10-10 for Q410; untouched here.

M2 `proofs/code-1014-q458/censusk.py`: no symmetric functions and no sympy. Uses Theorem 2.2 to replace the entire x-side by walk counts, and sums (9) directly.

Census (Theorem 2.2), three readings. (i) `mech2.py`: $[x^0]\dot{e}_r$ computed by exact Laurent dictionary arithmetic equals $\binom{n}{r/2}$ for all r, $1 \le n \le 8$; (ii) the same quantity by literal enumeration of all $\binom{2n}{r}$ subsets, $1 \le n \le 6$ — agrees; (iii) `verify_censusk.py`: the general-k statement (3) against brute Laurent multiplication of k elementary symmetric functions, all $r \in [0, 2n]^k$ for $1 \le n \le 5$, $k \in \{1, 2, 3\}$ — **2875 of 2875**, no mismatch. Also `walk.py`: the identity (4) holds as an exact polynomial identity for $k \le 5$, and $[z^r]L^n = W_n(r - n)$ in **697 of 697** checks.

Theorem 3.5. `verifyM1.py`: the full polynomial $c_{(}1^n)(t)$ from engine M1 equals (7) for $\ell \in \{0, 1, 2\}$ and $n = 1, \ldots, 8$ — **24 of 24**, *including the odd n, where both sides are* 0. `mech2.py`: engine M2 against (7) and against the degree law, $\ell \in \{0, 1, 2, 3\}$, $n = 1, \ldots, 12$ — **48 of 48**. Sample values ($\ell = 0$):

$$n = 2 :\ 1{-}t; \qquad n = 4 :\ t^3{-}3t{+}2; \qquad n = 6 :\ -t^6{+}5t^3{-}9t{+}5; \qquad n = 8 :\ t^{10}{-}7t^6{+}20t^3{-}28t{+}14.$$

The $\ell = 1$ degrees run $0, 2, 4, 4, 10, 14$ at $n = 2, 4, 6, 8, 10, 12$ — reproducing the $4 \to 10$ jump between $n = 8$ and $n = 10$ that the Q452 paper recorded as the sharpest row of its 9/9 table.

Cross-engine (M1 vs M2), general k. `xcheck_k.py`: full-polynomial agreement for $k \in \{1, 2\}$, $\ell \in \{0, 1\}$, $n \le 6$ (the limit of M1; the Q452 paper records M1 running out of memory at $k = 2, n = 7$). M2 reaches $k = 2, n = 28$ in 0.1 s.

Theorem 4.4. `thmB_fast.py`: for every (k, ℓ, n) with $1 \le k \le 4$, $0 \le \ell \le 3$, n even up to 36/30/20/14 (by k) *and* n in one of the two residue classes, the measured degree and the measured leading coefficient are compared against $n(n + 2k)/(4M)$ and ± 1 — **93 of 93**, no mismatch in either the exponent or the signed coefficient. Representative rows: $(k, \ell, n) = (1, 0, 34) \to t^{153}$ with coefficient -1; $(2, 0, 26) \to t^{65}$, -1; $(2, 2, 20) \to t^{24}$, $+1$; $(3, 0, 8) \to t^7$, $+1$; $(4, 3, 8) \to t^4$, -1. *A $k = 5$ arm was started and killed, not completed*: engine M2 raises $L_{1-k}(z)^n$ to the n-th power as an explicit dictionary, which at $k = 5$, $n = 14$ has up to 29^5 monomials. The verified range is $k \le 4$.

Negative control (must fail). Applying the $k = 1$ law of Theorem 3.5 verbatim at $k \geq 2$: it fails at $n = 2, 6, 8, 12, 14, 18, 20, 24, 26$ and holds at $n = 4, 10, 16, 22, 28$ for $(k, \ell) = (2, 0)$ — i.e. it holds exactly on the residue class where $N \mid n + 2$, and fails elsewhere. A $k = 1$ formula that matched at $k = 2$ would mean the derivation's hypothesis was not what it is claimed to be; it does not match, so the control fires.

Satisfiability / non-vacuity control. `topterm.py` prints, for each (k, ℓ, n), the top exponent, the top coefficient *as a number*, and the number of (y, σ) pairs attaining it. Across $k \in \{2, 3, 4\}$, $\ell \in \{0, 1\}$, $n \leq 20$: the top coefficient is never 0; it is exactly ± 1 precisely when a single (y, σ) attains the top — which happens precisely in the two residue classes of Theorem 4.4 — and is a larger integer $(3, 9, 15, 45, 105, 153, \dots)$ when several tie. This is the positive companion to the negative control: the kill in Corollary 4.6 is accompanied by an explicitly satisfiable witness at the top of every row.

7 What is proved, what is not

Proved. Theorem 2.2 (the census is a walk, all k, all n, fibre (1^n)); Theorem 3.5 (the exact $k = 1$ degree law, all even n, all ℓ — this is **Q458**, and it closes gap (G) of the Q452 paper); Proposition 3.7; Theorem 4.4 (exact degree and leading coefficient at every k on two residue classes — this is **Q457**); Corollary 4.6.

Not proved.

1. The exact degree for n outside the two residue classes at $k \geq 2$. Here the maximising vertex of Lemma 4.2 is unrealisable and the maximum passes to a non-vertex point of the ℓ^1 ball, where several (y, σ, ε) tie. Nothing in the argument breaks — the obstruction is only that Lemma 4.3's divisibility has no solution and the next-best p is not unique, so a cancellation analysis is needed that this paper does not perform. Measured leading coefficients there are nonzero in all 50 rows examined, but that is data, not a proof.

2. Anything inside Korff's window $h > 2$ *and* $n \leq w$. See §5(2). This is now, as far as the Q430/Q452/Q457/Q458 line is concerned, the whole of what remains.

3. Fibres other than (1^n).